

Full Derivation of the Expanded Welch-Satterthwaite Mutual Information Test

Two Independent Discrete Populations

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Textbook-style mathematical derivation

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Purpose

This chapter derives a deterministic test for comparing the mutual information of two independent discrete populations. The null hypothesis is

$$H_0 : I(P) = I(Q). \tag{1}$$

The construction follows the broad logic of Welch’s test:

1. estimate the difference between two population quantities;
2. divide that difference by its estimated standard error;
3. account for uncertainty in the estimated standard error by using a Student distribution with Satterthwaite effective degrees of freedom.

The important difference is that mutual information is a nonlinear functional of a complete joint probability table. Its variance estimator is therefore not an ordinary sample variance. The expanded method derives the first-order sampling uncertainty of that complete variance functional and uses it to calculate MI-specific component degrees of freedom.

The derivation below is written for natural logarithms, so MI is measured in nats.

1. Statistical Setting

Let

$$Z_1^{(P)}, \dots, Z_{n_P}^{(P)} \stackrel{\text{iid}}{\sim} P, \quad Z_1^{(Q)}, \dots, Z_{n_Q}^{(Q)} \stackrel{\text{iid}}{\sim} Q, \quad (2)$$

where every observation is a pair

$$Z = (X, Y) \in \{1, \dots, r\} \times \{1, \dots, c\}. \quad (3)$$

The two samples are independent. Write the population cell probabilities as

$$p_{ij} = \Pr_P(X = i, Y = j), \quad q_{ij} = \Pr_Q(X = i, Y = j). \quad (4)$$

For population P , define the marginal probabilities

$$p_{i+} = \sum_{j=1}^c p_{ij}, \quad p_{+j} = \sum_{i=1}^r p_{ij}. \quad (5)$$

The corresponding definitions apply to Q .

The regular derivation assumes a fixed finite alphabet and positive population support:

$$p_{ij} > 0, \quad q_{ij} > 0 \quad (6)$$

for all modelled cells. This makes the logarithms and derivatives below well-defined. Section 16 explains what changes when observed cells are empty or the population is at independence.

2. Mutual Information as a Functional

For a distribution P , define the local-information score of cell (i, j) by

$$\ell_P(i, j) = \log \left(\frac{p_{ij}}{p_{i+}p_{+j}} \right). \quad (7)$$

The population mutual information is

$$\begin{aligned} I(P) &= \sum_{i=1}^r \sum_{j=1}^c p_{ij} \log \left(\frac{p_{ij}}{p_{i+}p_{+j}} \right) \\ &= \mathbb{E}_P\{\ell_P(X, Y)\}. \end{aligned} \quad (8)$$

For notational convenience, write

$$\mu_P = I(P). \quad (9)$$

The subscript on ℓ_P matters. The score is not a fixed value attached to a cell independently of the distribution. If P changes, the joint probability, both relevant marginal probabilities, and therefore the score all change.

3. Plug-In Estimation

Let $N_{ij}^{(P)}$ be the observed count in cell (i, j) of the first table. Then

$$n_P = \sum_{i=1}^r \sum_{j=1}^c N_{ij}^{(P)}, \quad \hat{p}_{ij} = \frac{N_{ij}^{(P)}}{n_P}. \quad (10)$$

The empirical marginals are

$$\hat{p}_{i+} = \sum_{j=1}^c \hat{p}_{ij}, \quad \hat{p}_{+j} = \sum_{i=1}^r \hat{p}_{ij}. \quad (11)$$

The plug-in local-information score and plug-in MI are

$$\hat{\ell}_{ij} = \log \left(\frac{\hat{p}_{ij}}{\hat{p}_{i+} \hat{p}_{+j}} \right) \quad (12)$$

and

$$\hat{I}(P) = \sum_{i=1}^r \sum_{j=1}^c \hat{p}_{ij} \hat{\ell}_{ij}. \quad (13)$$

Equivalently, if $\hat{P}$ is the empirical distribution, then

$$\hat{I}(P) = I(\hat{P}). \quad (14)$$

The same calculations produce $\hat{I}(Q)$ from the second table.

4. Leading Bias Correction

4.1 Bias of plug-in entropy

For a discrete variable with k positive-probability categories, the leading bias of the plug-in entropy estimator is

$$\mathbb{E}(\hat{H}) - H = -\frac{k-1}{2n} + O(n^{-2}). \quad (15)$$

Mutual information can be written as

$$I(X; Y) = H(X) + H(Y) - H(X, Y). \quad (16)$$

Under full support, X has r categories, Y has c categories, and the joint variable (X, Y) has rc categories. Applying the entropy bias formula to these three terms gives

$$\begin{aligned} \text{Bias}(\hat{I}) &\approx -\frac{r-1}{2n} - \frac{c-1}{2n} - \left(-\frac{rc-1}{2n}\right) \\ &= \frac{rc - r - c + 1}{2n} \\ &= \frac{(r-1)(c-1)}{2n}. \end{aligned} \quad (17)$$

Define

$$d = (r-1)(c-1). \quad (18)$$

4.2 Bias-corrected MI difference

The leading-bias-corrected estimators are

$$\hat{I}_{\text{BC}}(P) = \hat{I}(P) - \frac{d}{2n_P} \quad (19)$$

and

$$\hat{I}_{\text{BC}}(Q) = \hat{I}(Q) - \frac{d}{2n_Q}. \quad (20)$$

The estimated population difference is

$$\hat{\Delta}_{\text{BC}} = \hat{I}_{\text{BC}}(P) - \hat{I}_{\text{BC}}(Q). \quad (21)$$

Expanding this expression gives

$$\hat{\Delta}_{\text{BC}} = \hat{I}(P) - \hat{I}(Q) - \frac{d}{2n_P} + \frac{d}{2n_Q}. \quad (22)$$

For fixed table dimensions and fixed sample sizes, the correction terms are constants. They change the estimated difference but do not add sampling variance and do not change the influence function derived next.

5. Deriving the Influence Function of MI

5.1 Contaminate one cell

Fix a cell $z = (x, y)$. Define a path of distributions

$$P_\varepsilon = (1 - \varepsilon)P + \varepsilon\delta_z, \quad (23)$$

where δ_z places probability one on cell z . At $\varepsilon = 0$ the distribution is P . Increasing ε moves a small amount of probability toward cell (x, y) while preserving total probability one.

For an arbitrary cell (i, j) ,

$$p_{ij}(\varepsilon) = (1 - \varepsilon)p_{ij} + \varepsilon\mathbf{1}\{i = x, j = y\}. \quad (24)$$

Therefore,

$$\left. \frac{d}{d\varepsilon} p_{ij}(\varepsilon) \right|_{\varepsilon=0} = \mathbf{1}\{i = x, j = y\} - p_{ij}. \quad (25)$$

Summing over columns gives

$$p_{i+}(\varepsilon) = (1 - \varepsilon)p_{i+} + \varepsilon\mathbf{1}\{i = x\}, \quad (26)$$

so

$$\left. \frac{d}{d\varepsilon} p_{i+}(\varepsilon) \right|_{\varepsilon=0} = \mathbf{1}\{i = x\} - p_{i+}. \quad (27)$$

Likewise,

$$\left. \frac{d}{d\varepsilon} p_{+j}(\varepsilon) \right|_{\varepsilon=0} = \mathbf{1}\{j = y\} - p_{+j}. \quad (28)$$

5.2 Differentiate the local-information score

Under P_ε , the score of cell (i, j) is

$$\ell_{P_\varepsilon}(i, j) = \log p_{ij}(\varepsilon) - \log p_{i+}(\varepsilon) - \log p_{+j}(\varepsilon). \quad (29)$$

Use

$$\frac{d}{d\varepsilon} \log u(\varepsilon) = \frac{u'(\varepsilon)}{u(\varepsilon)}. \quad (30)$$

The derivative of the joint-probability term at zero is

$$\begin{aligned} \left. \frac{d}{d\varepsilon} \log p_{ij}(\varepsilon) \right|_{\varepsilon=0} &= \frac{\mathbf{1}\{i = x, j = y\} - p_{ij}}{p_{ij}} \\ &= \frac{\mathbf{1}\{i = x, j = y\}}{p_{ij}} - 1. \end{aligned} \quad (31)$$

Similarly,

$$\left. \frac{d}{d\varepsilon} \log p_{i+}(\varepsilon) \right|_{\varepsilon=0} = \frac{\mathbf{1}\{i = x\}}{p_{i+}} - 1 \quad (32)$$

and

$$\left. \frac{d}{d\varepsilon} \log p_{+j}(\varepsilon) \right|_{\varepsilon=0} = \frac{\mathbf{1}\{j = y\}}{p_{+j}} - 1. \quad (33)$$

Subtracting the two marginal derivatives from the joint derivative gives

$$\dot{\ell}_P(i, j; z) = \frac{\mathbf{1}\{i = x, j = y\}}{p_{ij}} - \frac{\mathbf{1}\{i = x\}}{p_{i+}} - \frac{\mathbf{1}\{j = y\}}{p_{+j}} + 1, \quad (34)$$

where

$$\dot{\ell}_P(i, j; z) = \left. \frac{d}{d\varepsilon} \ell_{P_\varepsilon}(i, j) \right|_{\varepsilon=0}. \quad (35)$$

5.3 A derivative rule for distribution-dependent expectations

Suppose

$$F(P_\varepsilon) = \sum_{i,j} p_{ij}(\varepsilon) h_{P_\varepsilon}(i, j), \quad (36)$$

where both the probabilities and the values h_{P_ε} depend on ε . The product rule gives

$$\begin{aligned} \left. \frac{d}{d\varepsilon} F(P_\varepsilon) \right|_{\varepsilon=0} &= \sum_{i,j} [\mathbf{1}\{i = x, j = y\} - p_{ij}] h_P(i, j) \\ &\quad + \sum_{i,j} p_{ij} \dot{h}_P(i, j; z). \end{aligned} \quad (37)$$

The first sum is

$$h_P(x, y) - \mathbb{E}_P\{h_P(X, Y)\}. \quad (38)$$

Therefore,

$$\boxed{\mathbb{IF}_{F,P}(z) = h_P(z) - \mathbb{E}_P(h_P) + \mathbb{E}_P\{\dot{h}_P(Z; z)\}.} \quad (39)$$

The last term is essential whenever the quantity being averaged changes with the underlying distribution.

5.4 Apply the rule to MI

For MI, take

$$h_P(i, j) = \ell_P(i, j). \quad (40)$$

We first evaluate the expectation of the score derivative:

$$\begin{aligned}
\mathbb{E}_P\{\dot{\ell}_P(Z; z)\} &= \sum_{i,j} p_{ij} \left[\frac{\mathbf{1}\{i=x, j=y\}}{p_{ij}} - \frac{\mathbf{1}\{i=x\}}{p_{i+}} - \frac{\mathbf{1}\{j=y\}}{p_{+j}} + 1 \right] \\
&= 1 - \frac{\sum_j p_{xj}}{p_{x+}} - \frac{\sum_i p_{iy}}{p_{+y}} + \sum_{i,j} p_{ij} \\
&= 1 - 1 - 1 + 1 \\
&= 0.
\end{aligned} \tag{41}$$

The derivative rule therefore yields

$$\begin{aligned}
\text{IF}_{I,P}(x, y) &= \ell_P(x, y) - \mathbb{E}_P(\ell_P) + 0 \\
&= \ell_P(x, y) - \mu_P.
\end{aligned} \tag{42}$$

Define

$$\boxed{\psi_P(x, y) = \ell_P(x, y) - \mu_P.} \tag{43}$$

This is the influence function of mutual information.

5.5 First-order sampling variance of plug-in MI

The empirical distribution satisfies a first-order functional expansion:

$$I(\hat{P}) - I(P) = \frac{1}{n_P} \sum_{k=1}^{n_P} \psi_P(Z_k^{(P)}) + o_p(n_P^{-1/2}). \tag{44}$$

Because

$$\mathbb{E}_P\{\psi_P(Z)\} = 0, \tag{45}$$

the central limit theorem gives

$$\sqrt{n_P} \left\{ \hat{I}(P) - I(P) \right\} \xrightarrow{d} N\{0, V(P)\}, \tag{46}$$

where

$$\begin{aligned}
V(P) &= \text{Var}_P\{\psi_P(Z)\} \\
&= \mathbb{E}_P \left[\{\ell_P(Z) - \mu_P\}^2 \right].
\end{aligned} \tag{47}$$

Consequently,

$$\text{Var}\{\hat{I}(P)\} \approx \frac{V(P)}{n_P}. \tag{48}$$

The bias correction in Section 4 is a fixed constant at a given sample size, so it has the same first-order variance:

$$\text{Var}\{\hat{I}_{\text{BC}}(P)\} \approx \frac{V(P)}{n_P}. \tag{49}$$

6. The Two-Sample Standardized Statistic

Define the corresponding influence variance $V(Q)$ for the second population. Independence of the two samples implies

$$\text{Var}(\hat{\Delta}_{\text{BC}}) \approx \frac{V(P)}{n_P} + \frac{V(Q)}{n_Q}. \quad (50)$$

To estimate $V(P)$, define

$$m_{2,P} = \mathbb{E}_P\{\ell_P(Z)^2\}. \quad (51)$$

Since $\mu_P = \mathbb{E}_P(\ell_P)$,

$$V(P) = m_{2,P} - \mu_P^2. \quad (52)$$

Its plug-in estimate is

$$\begin{aligned} \hat{V}_P &= \sum_{i,j} \hat{p}_{ij} \left(\hat{\ell}_{ij} - \hat{I}(P) \right)^2 \\ &= \sum_{i,j} \hat{p}_{ij} \hat{\ell}_{ij}^2 - \hat{I}(P)^2. \end{aligned} \quad (53)$$

Define $\hat{V}_Q$ analogously. The two estimated contributions to the variance of the difference are

$$a = \frac{\hat{V}_P}{n_P}, \quad b = \frac{\hat{V}_Q}{n_Q}. \quad (54)$$

The estimated standard error is

$$\widehat{\text{SE}} = \sqrt{a + b} = \sqrt{\frac{\hat{V}_P}{n_P} + \frac{\hat{V}_Q}{n_Q}}. \quad (55)$$

The standardized statistic is

$$T = \frac{\hat{\Delta}_{\text{BC}}}{\sqrt{\hat{V}_P/n_P + \hat{V}_Q/n_Q}}. \quad (56)$$

If $V(P)$ and $V(Q)$ were known, a standard normal reference would follow from the two independent asymptotic expansions. In practice, both variances are estimated. Expanded Welch–Satterthwaite estimates the uncertainty in those variance estimates rather than treating the denominator as known.

7. Deriving the Influence Function of the MI Variance

7.1 The target variance functional

Recall that

$$V(P) = m_{2,P} - \mu_P^2, \quad (57)$$

where

$$m_{2,P} = \mathbb{E}_P\{\ell_P(Z)^2\}, \quad \mu_P = \mathbb{E}_P\{\ell_P(Z)\}. \quad (58)$$

The goal is to differentiate $V(P_\varepsilon)$ along the same contamination path used in Section 5. Define

$$g_P(x, y) = \left. \frac{d}{d\varepsilon} V(P_\varepsilon) \right|_{\varepsilon=0}. \quad (59)$$

This g_P is the influence function of the complete MI variance functional.

7.2 Differentiate the second moment

Under P_ε ,

$$m_2(P_\varepsilon) = \sum_{i,j} p_{ij}(\varepsilon) \ell_{P_\varepsilon}(i, j)^2. \quad (60)$$

Differentiate using the product rule:

$$\begin{aligned} \left. \frac{d}{d\varepsilon} m_2(P_\varepsilon) \right|_{\varepsilon=0} &= \sum_{i,j} [\mathbf{1}\{i = x, j = y\} - p_{ij}] \ell_P(i, j)^2 \\ &\quad + \sum_{i,j} p_{ij} \left. \frac{d}{d\varepsilon} \ell_{P_\varepsilon}(i, j)^2 \right|_{\varepsilon=0}. \end{aligned} \quad (61)$$

For the first sum,

$$\sum_{i,j} [\mathbf{1}\{i = x, j = y\} - p_{ij}] \ell_P(i, j)^2 = \ell_P(x, y)^2 - m_{2,P}. \quad (62)$$

For the second sum, the chain rule gives

$$\left. \frac{d}{d\varepsilon} \ell_{P_\varepsilon}(i, j)^2 \right|_{\varepsilon=0} = 2\ell_P(i, j) \dot{\ell}_P(i, j; z). \quad (63)$$

Therefore,

$$\text{IF}_{m_2,P}(x, y) = \ell_P(x, y)^2 - m_{2,P} + 2 \sum_{i,j} p_{ij} \ell_P(i, j) \dot{\ell}_P(i, j; z). \quad (64)$$

7.3 Evaluate the score-derivative sum

Substitute the expression for $\dot{\ell}_P$:

$$\begin{aligned} & \sum_{i,j} p_{ij} \ell_P(i,j) \dot{\ell}_P(i,j; z) \\ &= \sum_{i,j} p_{ij} \ell_P(i,j) \left[\frac{\mathbf{1}\{i = x, j = y\}}{p_{ij}} - \frac{\mathbf{1}\{i = x\}}{p_{i+}} - \frac{\mathbf{1}\{j = y\}}{p_{+j}} + 1 \right]. \end{aligned} \quad (65)$$

Evaluate the four terms separately.

The cell term is

$$\sum_{i,j} p_{ij} \ell_P(i,j) \frac{\mathbf{1}\{i = x, j = y\}}{p_{ij}} = \ell_P(x, y). \quad (66)$$

The row term is

$$\begin{aligned} \sum_{i,j} p_{ij} \ell_P(i,j) \frac{\mathbf{1}\{i = x\}}{p_{i+}} &= \frac{1}{p_{x+}} \sum_j p_{xj} \ell_P(x, j) \\ &= \mathbb{E}_P\{\ell_P(X, Y) \mid X = x\}. \end{aligned} \quad (67)$$

Define

$$R_P(x) = \mathbb{E}_P\{\ell_P(X, Y) \mid X = x\}. \quad (68)$$

The column term is

$$\begin{aligned} \sum_{i,j} p_{ij} \ell_P(i,j) \frac{\mathbf{1}\{j = y\}}{p_{+j}} &= \frac{1}{p_{+y}} \sum_i p_{iy} \ell_P(i, y) \\ &= \mathbb{E}_P\{\ell_P(X, Y) \mid Y = y\}. \end{aligned} \quad (69)$$

Define

$$C_P(y) = \mathbb{E}_P\{\ell_P(X, Y) \mid Y = y\}. \quad (70)$$

Finally, the constant term is

$$\sum_{i,j} p_{ij} \ell_P(i,j) = \mu_P. \quad (71)$$

Remembering the minus signs on the row and column terms,

$$\boxed{\sum_{i,j} p_{ij} \ell_P(i,j) \dot{\ell}_P(i,j; z) = \ell_P(x, y) - R_P(x) - C_P(y) + \mu_P.} \quad (72)$$

It follows that

$$\boxed{\begin{aligned} \text{IF}_{m_2, P}(x, y) &= \ell_P(x, y)^2 - m_{2, P} \\ &\quad + 2\{\ell_P(x, y) - R_P(x) - C_P(y) + \mu_P\}. \end{aligned}} \quad (73)$$

7.4 Differentiate the squared mean

From Section 5,

$$\text{IF}_{\mu,P}(x, y) = \ell_P(x, y) - \mu_P. \quad (74)$$

The ordinary chain rule for u^2 gives

$$\left. \frac{d}{d\varepsilon} \mu(P_\varepsilon)^2 \right|_{\varepsilon=0} = 2\mu_P \text{IF}_{\mu,P}(x, y). \quad (75)$$

Therefore,

$$\boxed{\text{IF}_{\mu^2,P}(x, y) = 2\mu_P \{\ell_P(x, y) - \mu_P\}.} \quad (76)$$

7.5 Combine the derivatives

Because

$$V(P) = m_{2,P} - \mu_P^2, \quad (77)$$

linearity of differentiation gives

$$g_P(x, y) = \text{IF}_{m_{2,P}}(x, y) - \text{IF}_{\mu^2,P}(x, y). \quad (78)$$

Substituting the two expressions above yields

$$\boxed{\begin{aligned} g_P(x, y) &= \ell_P(x, y)^2 - m_{2,P} \\ &\quad + 2\{\ell_P(x, y) - R_P(x) - C_P(y) + \mu_P\} \\ &\quad - 2\mu_P\{\ell_P(x, y) - \mu_P\}. \end{aligned}} \quad (79)$$

This is the complete variance-influence formula used by the expanded method. No local score is treated as fixed: the middle line accounts for changes to the joint cell, its row marginal, its column marginal, and the overall MI.

7.6 Verify that the influence function has mean zero

An influence function should be centred under its defining distribution. We can verify this directly.

For the first line,

$$\mathbb{E}_P\{\ell_P(Z)^2 - m_{2,P}\} = m_{2,P} - m_{2,P} = 0. \quad (80)$$

For the middle line, the law of iterated expectations gives

$$\mathbb{E}_P\{R_P(X)\} = \mathbb{E}_P[\mathbb{E}_P\{\ell_P(Z) \mid X\}] = \mu_P \quad (81)$$

and

$$E_P\{C_P(Y)\} = \mu_P. \quad (82)$$

Hence,

$$\begin{aligned} E_P\{\ell_P(Z) - R_P(X) - C_P(Y) + \mu_P\} &= \mu_P - \mu_P - \mu_P + \mu_P \\ &= 0. \end{aligned} \quad (83)$$

For the final line,

$$E_P[-2\mu_P\{\ell_P(Z) - \mu_P\}] = -2\mu_P(\mu_P - \mu_P) = 0. \quad (84)$$

Combining the three results,

$$\boxed{E_P\{g_P(X, Y)\} = 0.} \quad (85)$$

8. Sampling Uncertainty of the Estimated MI Variance

The plug-in variance estimator can be written as the empirical functional

$$\hat{V}_P = V(\hat{P}). \quad (86)$$

Because g_P is the influence function of V , its first-order expansion is

$$V(\hat{P}) - V(P) = \frac{1}{n_P} \sum_{k=1}^{n_P} g_P(Z_k^{(P)}) + o_p(n_P^{-1/2}). \quad (87)$$

Define

$$\tau_P^2 = \text{Var}_P\{g_P(X, Y)\}. \quad (88)$$

Since $E_P(g_P) = 0$, this can also be written as

$$\tau_P^2 = E_P\{g_P(X, Y)^2\}. \quad (89)$$

Taking the variance of the leading term in the expansion gives

$$\begin{aligned} \text{Var}\{\hat{V}_P\} &\approx \text{Var}\left\{\frac{1}{n_P} \sum_{k=1}^{n_P} g_P(Z_k^{(P)})\right\} \\ &= \frac{1}{n_P^2} \sum_{k=1}^{n_P} \text{Var}_P\{g_P(Z_k^{(P)})\} \\ &= \frac{1}{n_P^2} (n_P \tau_P^2) \\ &= \boxed{\frac{\tau_P^2}{n_P}}. \end{aligned} \quad (90)$$

The covariance terms vanish because observations within the sample are independent.

This result answers the key question: τ_P^2/n_P is the first-order sampling variance of the estimated MI influence variance $\hat{V}_P$.

9. Estimate the Variance-Influence Quantities

All population quantities are replaced by their empirical versions. For the first table, calculate

$$\hat{\mu}_P = \sum_{i,j} \hat{p}_{ij} \hat{\ell}_{ij}, \quad (91)$$

$$\hat{m}_{2,P} = \sum_{i,j} \hat{p}_{ij} \hat{\ell}_{ij}^2, \quad (92)$$

and

$$\hat{V}_P = \hat{m}_{2,P} - \hat{\mu}_P^2. \quad (93)$$

For each nonempty row and column, calculate

$$\hat{R}_P(i) = \frac{\sum_j \hat{p}_{ij} \hat{\ell}_{ij}}{\hat{p}_{i+}} \quad (94)$$

and

$$\hat{C}_P(j) = \frac{\sum_i \hat{p}_{ij} \hat{\ell}_{ij}}{\hat{p}_{+j}}. \quad (95)$$

For every cell, evaluate

$$\begin{aligned} \hat{g}_P(i, j) = & \hat{\ell}_{ij}^2 - \hat{m}_{2,P} \\ & + 2\{\hat{\ell}_{ij} - \hat{R}_P(i) - \hat{C}_P(j) + \hat{\mu}_P\} \\ & - 2\hat{\mu}_P\{\hat{\ell}_{ij} - \hat{\mu}_P\}. \end{aligned} \quad (96)$$

In exact arithmetic the weighted mean of $\hat{g}_P$ is zero. The implementation nevertheless centres it explicitly for numerical stability:

$$\bar{g}_P = \sum_{i,j} \hat{p}_{ij} \hat{g}_P(i, j). \quad (97)$$

Then

$$\boxed{\hat{\tau}_P^2 = \sum_{i,j} \hat{p}_{ij} \{\hat{g}_P(i, j) - \bar{g}_P\}^2.} \quad (98)$$

Repeat the complete calculation for table Q to obtain $\hat{V}_Q$ and $\hat{\tau}_Q^2$.

10. Satterthwaite Moment Matching for One Variance Component

10.1 The scaled chi-squared approximation

A chi-squared random variable $U \sim \chi_\nu^2$ has

$$\mathbb{E}(U) = \nu, \quad \text{Var}(U) = 2\nu. \quad (99)$$

Therefore,

$$Y = m \frac{U}{\nu} \quad (100)$$

has

$$\mathbb{E}(Y) = \frac{m}{\nu} \mathbb{E}(U) = m \quad (101)$$

and

$$\begin{aligned} \text{Var}(Y) &= \frac{m^2}{\nu^2} \text{Var}(U) \\ &= \frac{m^2}{\nu^2} (2\nu) \\ &= \frac{2m^2}{\nu}. \end{aligned} \quad (102)$$

Solving the last equation for ν gives the general Satterthwaite moment-matching rule

$$\boxed{\nu = \frac{2m^2}{\text{Var}(Y)}}. \quad (103)$$

10.2 Apply the rule to $\widehat{V}_P$

Approximate the positive variance estimator by

$$\widehat{V}_P \sim V(P) \frac{\chi_{\nu_{V,P}}^2}{\nu_{V,P}}. \quad (104)$$

The scaled chi-squared approximation has

$$\mathbb{E}(\widehat{V}_P) \approx V(P) \quad (105)$$

and

$$\text{Var}(\widehat{V}_P) \approx \frac{2V(P)^2}{\nu_{V,P}}. \quad (106)$$

Section 8 independently derived

$$\text{Var}(\widehat{V}_P) \approx \frac{\tau_P^2}{n_P}. \quad (107)$$

Equating the two approximations gives

$$\frac{2V(P)^2}{\nu_{V,P}} = \frac{\tau_P^2}{n_P}. \quad (108)$$

Multiply both sides by $n_P \nu_{V,P}$:

$$2n_P V(P)^2 = \nu_{V,P} \tau_P^2. \quad (109)$$

Divide by τ_P^2 :

$$\boxed{\nu_{V,P} = \frac{2n_P V(P)^2}{\tau_P^2}.} \quad (110)$$

The empirical component degrees of freedom are therefore

$$\boxed{\widehat{\nu}_{V,P} = \frac{2n_P \widehat{V}_P^2}{\widehat{\tau}_P^2}.} \quad (111)$$

Likewise,

$$\boxed{\widehat{\nu}_{V,Q} = \frac{2n_Q \widehat{V}_Q^2}{\widehat{\tau}_Q^2}.} \quad (112)$$

10.3 Why dividing by sample size does not change component degrees of freedom

The denominator of T uses

$$A = \frac{\widehat{V}_P}{n_P}, \quad (113)$$

not $\widehat{V}_P$ itself. From the scaled chi-squared approximation,

$$A \simeq \frac{V(P)}{n_P} \frac{\chi_{\nu_{V,P}}^2}{\nu_{V,P}}. \quad (114)$$

Only the scale has changed. The chi-squared degrees of freedom remain $\nu_{V,P}$. The same argument applies to

$$B = \frac{\widehat{V}_Q}{n_Q}. \quad (115)$$

11. Combine the Two Variance Components

The estimated squared standard error is

$$S^2 = A + B = \frac{\hat{V}_P}{n_P} + \frac{\hat{V}_Q}{n_Q}. \quad (116)$$

Let the approximate means of the two components be

$$a_0 = \frac{V(P)}{n_P}, \quad b_0 = \frac{V(Q)}{n_Q}. \quad (117)$$

Under the component approximations,

$$\text{Var}(A) \approx \frac{2a_0^2}{\nu_{V,P}} \quad (118)$$

and

$$\text{Var}(B) \approx \frac{2b_0^2}{\nu_{V,Q}}. \quad (119)$$

Because the P and Q samples are independent, A and B are independent. Thus,

$$\begin{aligned} \text{Var}(S^2) &= \text{Var}(A + B) \\ &= \text{Var}(A) + \text{Var}(B) \\ &\approx \frac{2a_0^2}{\nu_{V,P}} + \frac{2b_0^2}{\nu_{V,Q}}. \end{aligned} \quad (120)$$

Now approximate the sum by one scaled chi-squared variable:

$$S^2 \sim (a_0 + b_0) \frac{\chi_\nu^2}{\nu}. \quad (121)$$

This approximation has variance

$$\text{Var}(S^2) \approx \frac{2(a_0 + b_0)^2}{\nu}. \quad (122)$$

Match the two variance expressions:

$$\frac{2(a_0 + b_0)^2}{\nu} = \frac{2a_0^2}{\nu_{V,P}} + \frac{2b_0^2}{\nu_{V,Q}}. \quad (123)$$

Cancel the factor of two:

$$\frac{(a_0 + b_0)^2}{\nu} = \frac{a_0^2}{\nu_{V,P}} + \frac{b_0^2}{\nu_{V,Q}}. \quad (124)$$

Solve for ν :

$$\nu = \frac{(a_0 + b_0)^2}{a_0^2/\nu_{V,P} + b_0^2/\nu_{V,Q}}. \quad (125)$$

Finally, replace the population components and component degrees of freedom by their estimates:

$$a = \frac{\hat{V}_P}{n_P}, \quad b = \frac{\hat{V}_Q}{n_Q}. \quad (126)$$

The expanded Welch-Satterthwaite degrees of freedom are

$$\boxed{\hat{\nu}_{\text{expanded}} = \frac{(a + b)^2}{a^2/\hat{\nu}_{V,P} + b^2/\hat{\nu}_{V,Q}}}. \quad (127)$$

12. The Final Reference Distribution and P-Value

The expanded method compares the observed statistic

$$T = \frac{\hat{\Delta}_{\text{BC}}}{\sqrt{a+b}} \quad (128)$$

with a Student distribution having $\hat{\nu}_{\text{expanded}}$ degrees of freedom. The two-sided p-value is

$$p_{\text{expanded}} = 2 \left[1 - F_{t_{\hat{\nu}_{\text{expanded}}}}(|T|) \right], \quad (129)$$

where $F_{t_{\nu}}$ is the cumulative distribution function of a Student random variable with ν degrees of freedom.

At significance level α , reject $H_0 : I(P) = I(Q)$ when

$$p_{\text{expanded}} < \alpha. \quad (130)$$

Equivalently, reject when

$$|T| > t_{1-\alpha/2, \hat{\nu}_{\text{expanded}}}. \quad (131)$$

13. Complete Calculation from Two Count Tables

For clarity, the entire procedure is collected here without omitting any intermediate quantity.

Step 1: Convert counts to probabilities

For each group $G \in \{P, Q\}$,

$$\hat{p}_{ij}^{(G)} = \frac{N_{ij}^{(G)}}{n_G}. \quad (132)$$

Step 2: Calculate row and column marginals

$$\hat{p}_{i+}^{(G)} = \sum_j \hat{p}_{ij}^{(G)}, \quad \hat{p}_{+j}^{(G)} = \sum_i \hat{p}_{ij}^{(G)}. \quad (133)$$

Step 3: Calculate local-information scores

$$\hat{\ell}_{ij}^{(G)} = \log \left(\frac{\hat{p}_{ij}^{(G)}}{\hat{p}_{i+}^{(G)} \hat{p}_{+j}^{(G)}} \right). \quad (134)$$

Step 4: Calculate plug-in MI

$$\hat{I}(G) = \sum_{i,j} \hat{p}_{ij}^{(G)} \hat{\ell}_{ij}^{(G)}. \quad (135)$$

Step 5: Correct the leading bias

$$\hat{I}_{\text{BC}}(G) = \hat{I}(G) - \frac{(r-1)(c-1)}{2n_G}. \quad (136)$$

Step 6: Form the estimated difference

$$\hat{\Delta}_{\text{BC}} = \hat{I}_{\text{BC}}(P) - \hat{I}_{\text{BC}}(Q). \quad (137)$$

Step 7: Calculate the first two score moments

For each group,

$$\hat{\mu}_G = \sum_{i,j} \hat{p}_{ij}^{(G)} \hat{\ell}_{ij}^{(G)}, \quad (138)$$

$$\hat{m}_{2,G} = \sum_{i,j} \hat{p}_{ij}^{(G)} \left(\hat{\ell}_{ij}^{(G)} \right)^2, \quad (139)$$

and

$$\hat{V}_G = \hat{m}_{2,G} - \hat{\mu}_G^2. \quad (140)$$

Step 8: Calculate conditional score means

$$\hat{R}_G(i) = \frac{\sum_j \hat{p}_{ij}^{(G)} \hat{\ell}_{ij}^{(G)}}{\hat{p}_{i+}^{(G)}}, \quad (141)$$

$$\hat{C}_G(j) = \frac{\sum_i \hat{p}_{ij}^{(G)} \hat{\ell}_{ij}^{(G)}}{\hat{p}_{+j}^{(G)}}. \quad (142)$$

Step 9: Calculate the variance influence in each cell

$$\begin{aligned} \hat{g}_G(i, j) = & \left(\hat{\ell}_{ij}^{(G)} \right)^2 - \hat{m}_{2,G} \\ & + 2 \left\{ \hat{\ell}_{ij}^{(G)} - \hat{R}_G(i) - \hat{C}_G(j) + \hat{\mu}_G \right\} \\ & - 2 \hat{\mu}_G \left\{ \hat{\ell}_{ij}^{(G)} - \hat{\mu}_G \right\}. \end{aligned} \quad (143)$$

Step 10: Calculate variance-influence variability

$$\bar{g}_G = \sum_{i,j} \hat{p}_{ij}^{(G)} \hat{g}_G(i, j), \quad (144)$$

$$\hat{\tau}_G^2 = \sum_{i,j} \hat{p}_{ij}^{(G)} \{ \hat{g}_G(i, j) - \bar{g}_G \}^2. \quad (145)$$

Step 11: Calculate component degrees of freedom

$$\hat{\nu}_{V,G} = \frac{2n_G \hat{V}_G^2}{\hat{\tau}_G^2}. \quad (146)$$

Step 12: Calculate the two standard-error components

$$a = \frac{\hat{V}_P}{n_P}, \quad b = \frac{\hat{V}_Q}{n_Q}. \quad (147)$$

Step 13: Calculate the standard error and statistic

$$\widehat{\text{SE}} = \sqrt{a + b}, \quad (148)$$

$$T = \frac{\hat{\Delta}_{\text{BC}}}{\widehat{\text{SE}}}. \quad (149)$$

Step 14: Combine the component degrees of freedom

$$\hat{\nu}_{\text{expanded}} = \frac{(a + b)^2}{a^2/\hat{\nu}_{V,P} + b^2/\hat{\nu}_{V,Q}}. \quad (150)$$

Step 15: Calculate the two-sided p-value

$$p_{\text{expanded}} = 2 \left[1 - F_{t_{\hat{\nu}_{\text{expanded}}}}(|T|) \right]. \quad (151)$$

14. Interpretation of the Effective Degrees of Freedom

The component formula is

$$\hat{\nu}_{V,P} = \frac{2n_P \hat{V}_P^2}{\hat{\tau}_P^2}. \quad (152)$$

Its numerator contains sample size and the squared magnitude of the MI variance. Its denominator measures how strongly that estimated variance changes when probability mass moves among cells.

If the cell sensitivities $\hat{g}_P(i, j)$ are similar, then $\hat{\tau}_P^2$ is small. The estimated variance is relatively stable, the component degrees of freedom are large, and the Student reference is close to a normal reference.

If a small number of cells have unusually large sensitivities, then $\hat{\tau}_P^2$ is large. The estimated variance is less stable, the component degrees of freedom are smaller, and the Student reference develops heavier tails. For a fixed observed $|T|$, heavier tails produce a larger, more cautious p-value.

This mechanism is table-dependent. Unlike the simple assignment $\nu_{V,P} = n_P - 1$, it can respond to skewness, sparse empirical support, and unequal influence among cells.

15. Relationship to Ordinary Welch-Satterthwaite

An ordinary Welch test uses sample variances whose exact scaled chi-squared distributions are available under normal sampling. A conventional component with sample size n_P is assigned $n_P - 1$ degrees of freedom, leading to

$$\hat{\nu}_{\text{simple}} = \frac{(a + b)^2}{a^2/(n_P - 1) + b^2/(n_Q - 1)}. \quad (153)$$

Expanded Welch keeps the same Satterthwaite combination but replaces the ordinary component degrees of freedom with quantities derived from the MI variance functional:

$$n_P - 1 \quad \longrightarrow \quad \hat{\nu}_{V,P} = \frac{2n_P \hat{V}_P^2}{\hat{\tau}_P^2}, \quad (154)$$

and similarly for Q .

Thus, the expansion is not a different standard error and not a second bias correction. It is an MI-specific calculation of how uncertain each estimated variance component is.

16. Regularity Conditions and Limits of the Derivation

16.1 Fixed finite alphabet

The asymptotic expansions assume that r and c remain fixed as the sample sizes increase. The derivation does not establish validity when the alphabet grows with sample size.

16.2 Positive population support

The derivatives of $\log p_{ij}$ require positive population probabilities. The mathematical derivation therefore applies to cells in the fixed positive support of the population. Structural-zero models require a separate support definition and a corresponding bias dimension.

In computation, an observed zero-count cell contributes zero to probability-weighted sums under the convention $0 \log 0 = 0$. This keeps the plug-in calculation finite, but it does not remove the underlying smoothness assumption. Very sparse tables remain a finite-sample stress regime rather than a setting in which the approximation becomes exact.

16.3 Nondegenerate first-order MI variance

At exact independence,

$$p_{ij} = p_{i+}p_{+j}, \quad (155)$$

so

$$\ell_P(i, j) = 0, \quad \psi_P(i, j) = 0, \quad V(P) = 0. \quad (156)$$

The first-order normal approximation then degenerates. The statistic in this chapter requires a positive combined first-order variance and is intended for regular differential-MI comparisons away from this degeneracy. A test whose null is independence requires second-order theory and is not supplied by this derivation.

16.4 Independent samples

The variance addition and the Satterthwaite combination use independence of the P and Q samples. Paired, clustered, repeated-measures, or otherwise dependent samples require covariance terms and a different derivation.

16.5 Fixed table dimensions in the bias correction

The implemented leading correction uses

$$d = (r - 1)(c - 1) \quad (157)$$

from the configured table dimensions. If categories are selected or removed after observing the data, the nominal bias calculation and the sampling analysis can change.

16.6 The Student reference is an approximation

The derivation establishes first-order moments for the variance estimator and uses Satterthwaite moment matching. It does not prove that $\hat{V}_P$ is exactly scaled chi-squared. It also does not make the MI numerator exactly independent of its estimated denominator.

Consequently,

$$T \not\equiv t_{\hat{\nu}_{\text{expanded}}} \quad (158)$$

as an exact finite-sample identity. The Student distribution is a calibrated working reference motivated by the derived variance uncertainty. Its finite-sample accuracy must be established empirically.

16.7 Numerical validity conditions

The calculation requires finite positive values for

$$\hat{V}_P + \hat{V}_Q, \quad \hat{\tau}_P^2, \quad \hat{\tau}_Q^2, \quad (159)$$

and for the resulting component and combined degrees of freedom. If these conditions fail, the implementation reports the expanded result as invalid rather than manufacturing a p-value.

17. Computational Complexity

Each quantity is obtained by a fixed number of operations over an $r \times c$ table:

- cell probabilities and local scores require $O(rc)$ work;
- row and column reductions require $O(rc)$ work;
- $\hat{V}$, $\hat{g}$, and $\hat{\tau}^2$ require $O(rc)$ work;
- combining the two components requires constant work.

Therefore, the complete two-table method has

$$\boxed{\text{time complexity } O(rc), \quad \text{memory complexity } O(rc).} \quad (160)$$

No permutations, bootstrap samples, or Monte Carlo tables are required.

18. Correspondence with the Implementation

The implementation is in `src/welch_differential_mi/welch.py`. Its main quantities correspond to the derivation as follows.

Mathematical quantity	Implementation name
$\hat{I}$	<code>plugin_mi(...)</code>
d	<code>mi_df</code>
$\hat{\Delta}_{BC}$	<code>delta</code>
$\hat{V}_P, \hat{V}_Q$	<code>variance_p, variance_q</code>
a, b	<code>component_p, component_q</code>
T	<code>statistic</code>
$\hat{R}(i)$	<code>row_score_mean</code>
$\hat{C}(j)$	<code>column_score_mean</code>
$\hat{g}(i, j)$	<code>variance_influence</code>
$\hat{\tau}^2$	<code>influence_variance</code> inside <code>_variance_influence_component_df</code>
$\hat{\nu}_{V,P}, \hat{\nu}_{V,Q}$	<code>expanded_df_p, expanded_df_q</code>
$\hat{\nu}_{\text{expanded}}$	<code>expanded_df</code>
p_{expanded}	<code>expanded_p</code>

19. Derivation in One Chain

The complete mathematical chain is

$$P \longrightarrow \ell_P \longrightarrow I(P) = \mu_P \longrightarrow \psi_P = \ell_P - \mu_P \longrightarrow V(P) = \text{Var}_P(\psi_P) \quad (161)$$

followed by

$$V(P) \longrightarrow g_P = \text{IF}_{V,P} \longrightarrow \tau_P^2 = \text{Var}_P(g_P) \longrightarrow \nu_{V,P} = \frac{2n_P V(P)^2}{\tau_P^2}. \quad (162)$$

After repeating the chain for Q ,

$$(\nu_{V,P}, \nu_{V,Q}) \longrightarrow \nu_{\text{expanded}} \longrightarrow t_{\nu_{\text{expanded}}} \longrightarrow p_{\text{expanded}}. \quad (163)$$

The method therefore uses one influence function to estimate the sampling variance of MI and a second influence function to estimate the sampling uncertainty of that variance. Satterthwaite moment matching converts the second uncertainty calculation into the effective degrees of freedom used to interpret the standardized MI difference.

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